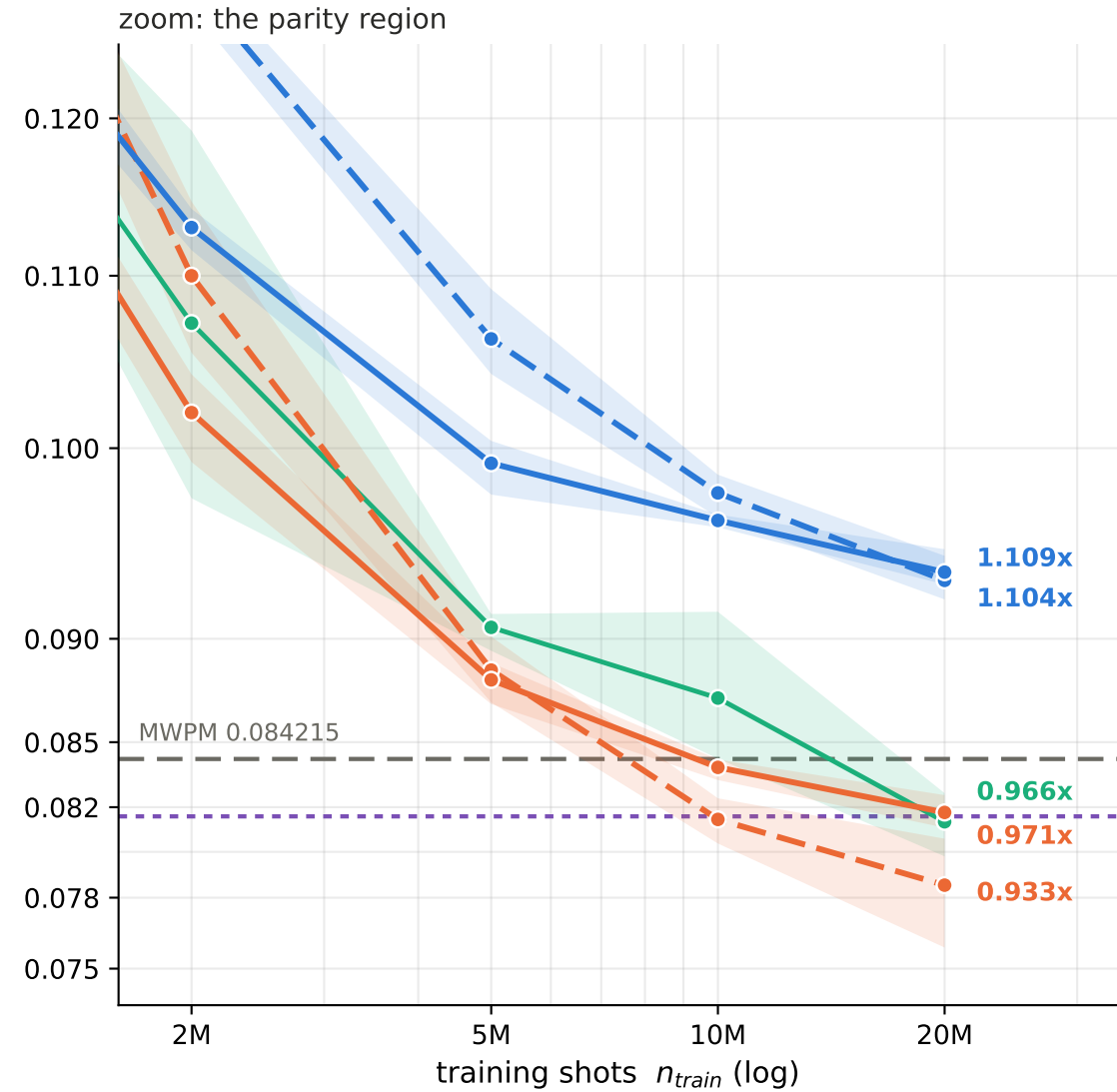
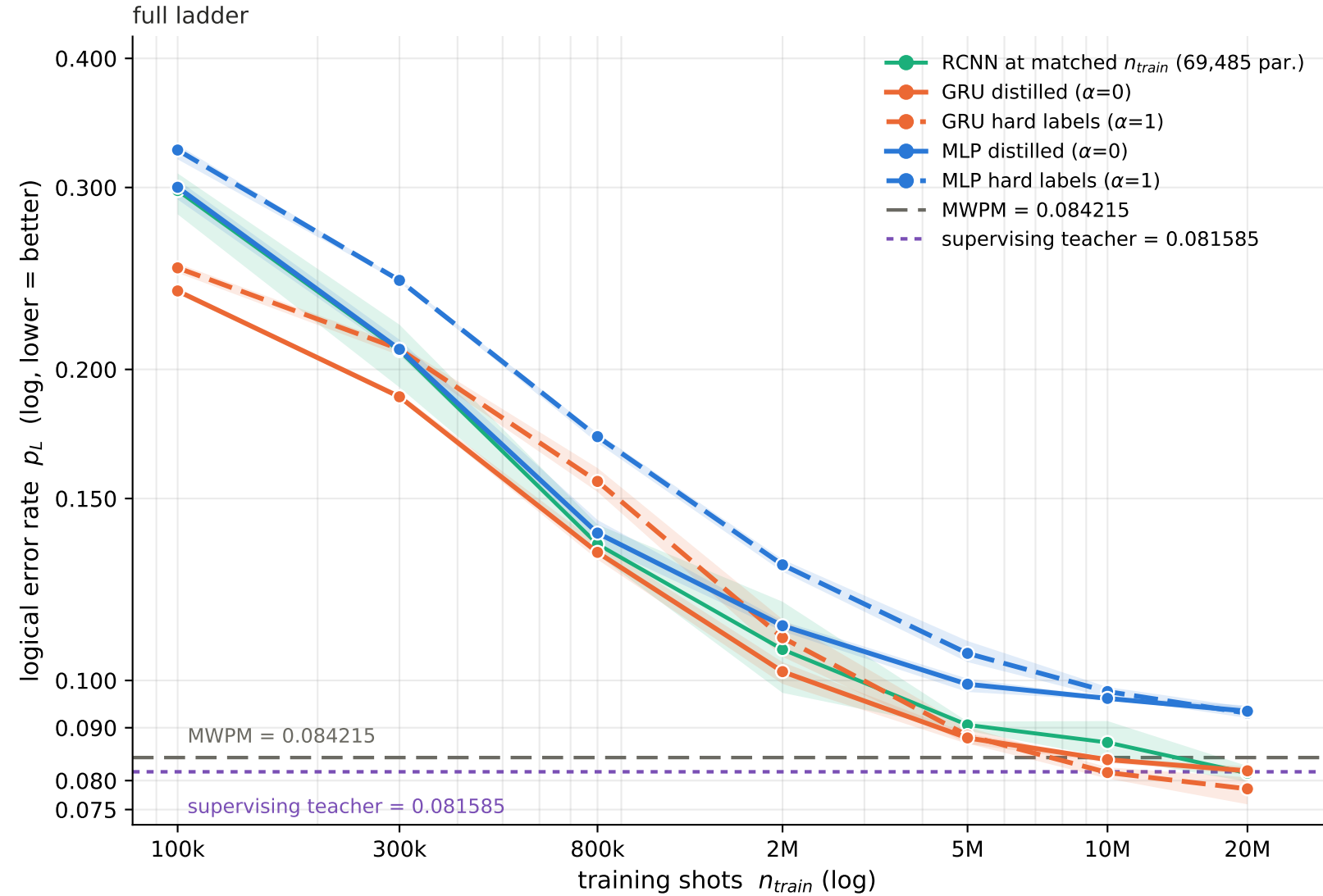


Experiment 9 — distillation vs hard labels at $r=5$

Surface code $d=5$, $p=0.01$, $r=5$ · 200,000-shot validation partition [20.0M, 20.2M) · 3 seeds per rung per arm · band = full range



Solid = distilled, dashed = hard labels, following Experiment 7. The purple line is the single frozen checkpoint that produced every soft target (seed 0 @ 20M); the green curve is the RCNN retrained at each n_{train} , a baseline that supervised nothing. Only a curve crossing below the purple line has beaten the model that taught it. In the zoom panel the RCNN, the distilled GRU and the supervisor all sit within 0.0005 of each other at 20M -- a tie, not an ordering. $T=1$, $lr=3e-3$, identical to the hard-label arm; sealed test [20.2M, 20.4M) not read.